We sincerely thank the reviewers for the detailed suggestions and positive feedback. All acknowledge IRIS's value as a real-world physics benchmark. **MCJw** acknowledges our "high quality dataset", "novel benchmark settings ... through multi-body collisions", and "explicit diagnostic systematic failure modes". **PXn6** finds the direction "meaningful and timely" and that IRIS "extends prior real-world datasets ... and provides a cleaner benchmarking protocol". **bRfX** ranks IRIS "quite highly in significance and timeliness". IRIS is the **first real-world benchmark with multi-body dynamics**. All clarifications below will be incorporated in camera-ready. **Primary IRIS results and multi-step findings (bRfX, MCJw):** Tab. 1 consolidates the prose results of §6.3 (per **bRfX**'s W1) and answers **MCJw**'s l.21 question on multi-step novelty: rollouts are known in dynamics modeling [38,39], but their systematic evaluation for *latent-space physical parameter* estimation is new, with both findings (single-body gain, multi-body divergence at $K>1$) exposed for the first time on real video. **Dataset scope, novelty over Delfys75 (bRfX, PXn6):** IRIS does *not* merely rescale Delfys75; main Tab. 3 documents advances on every axis: (i) *Duration* up to 150 s vs. ∼5 s; (ii) *Resolution* 4× pixel density; (iii) adds rotating cone and nonlinear pendulum to 90°; (iv) 3 entirely novel multi-body scenarios absent from any prior real-world benchmark; (v) 10 vs. 5 trials. The 65–73% VLM accuracy and CNN distribution-dependence are central *diagnostic findings*, not shortcomings. **Datasets/Benchmarks evaluation rubric (PXn6):** **PXn6** selected contribution type **Datasets/Benchmarks** (*"Benchmark as suggested in the title"*) and explicitly recognises IRIS's value: the direction is *"meaningful and timely"*; IRIS *"extends prior real-world datasets beyond single-body settings ... and provides a cleaner benchmarking protocol"*; the paper *"successfully exposes several important failure modes"*; and *"the dataset itself may become useful"*. **Yet PXn6**'s stated grounds for Borderline Reject are precisely *"the methodological component is not yet strong enough"* and the absence of *"a new, strong method that substantially solves the problem"*. The ECCV 2026 Reviewer Guidelines [1] for this contribution type state explicitly: **Do not penalize these submissions for: Limited methodological novelty**; acceptance is *"based primarily on: Whether the dataset/benchmark will provide clear value to the ECCV community"*. The criterion driving **PXn6**'s BR is the single criterion the rubric explicitly forbids penalizing; we hope **PXn6** revisits the rating under the rubric they themselves selected. **Reviewer MCJw: Bug nature (l.93,138):** *Conceptual*, not a software typo. The Euler step in [7] writes $z_{t+1}{=}z_t{+}\Delta t\,\dot{z}_t$, omitting the $\Delta t^2\,\ddot{z}_t$ term. Since $\ddot{z}_t$ is the only term depending on physics parameters γ, the gradient path is severed and parameters are unlearnable. We discovered this by running [7]'s code on IRIS and observing $G_\gamma^{(e)}{\approx}0$ throughout training (main Tab. 6); the authors of [7] have been notified. **Clarity (l.206, 245–259, 306, 337, Tabs. 4–8):** We will (i) define every symbol on first use ($f, g, \alpha, \mu, \kappa_{ij}, \zeta, N, N_f$); (ii) delineate at l.337 what is from [7] vs. new; (iii) add "so what?" takeaways to Tabs. 4–8 (see Tab. 1). **High-fidelity/Resolution/trials (l.325)**: 4K's pixel density reduces position quantization error by 4×, enabling metric-scale recovery 1080p cannot reliably achieve; 10 trials enable per-setting 95% uncertainty bounds. Tab. 2 confirms this (4K vs. downsampled 1080p). **Reviewer PXn6:**

Table 1. Parameter MAE on IRIS for +Corrected (C) vs. +Multi-step (M). Δ=M−C; **bold**=lower MAE. Green: M better. Red: M worse or diverges (†). **Takeaway:** +C is the reliable baseline; +M helps single-body stiffness-dominated phenomena but fails on multi-body, an open problem [38,39] IRIS exposes for the first time on real video.

Phenomenon	Setting	Param	C	M	Δ
Rotating cone	slow/mid/fast	α	C	**better**	−0.64/−1.21/−2.11
	fast	β	**C**	worse	+0.39
Pendulum	20°	L [m]	C	**better**	−0.44
	90°	L [m]	**C**	†	+20.49
Drop/fall ball	all	g [m/s^2]	**C**	worse	>0
Two mov. pend.	20°	L_i [m]	**1.1–3.0**	†	+741.1

Scenario	MAE_{4K}	MAE_{1080p}	Δ
Dropping_ball	**4.22**	4.45	+5.2%
Pendulum	**4.16**	4.73	+12.1%
Two_Moving_Pend(1S)	**4.40**	4.73	+7.1%

Table 2. MAE and resolution gain across IRIS scenarios with scalar GT (4K vs. downsampled 1080p).

Train-per-clip is per-instance fitting: Per-clip is deliberate identifiability evaluation [15], not a limitation. To test cross-clip generalization we ran a **multi-clip experiment** (80/20 split per phenomenon): multi-clip outperforms per-clip for linear phenomena but fails on the nonlinear pendulum, exactly the open problem IRIS surfaces (Tab. 3, left). **Fitted GT and simplified contact:** Fitted GT is documented via `measurement_type` in `parameters.json` with uncertainty. Continuous κ_{ij} is an acknowledged approximation (§3.3, §7); to confirm it is not fitting noise we compared κ=learned vs. κ=0: learned κ reduces reconstruction MSE by **6–19%** (Tab. 3, right). **Reviewer bRfX: Primary task evaluation (W1):** Tab. 1

Multi-clip vs. per-clip MAE (80/20)

Phenomenon	Param	Per	Multi	Δ
Dropping ball	g	0.207	**0.046**	−0.161
Falling ball	g	0.246	**0.166**	−0.080
Rotation	stiff.	1.037	**0.577**	−0.459
Rotation	damp.	**0.657**	0.851	+0.194
Pendulum	L	**0.507**	22.06	+21.55

Hit.-cones MSE (κ)

Set.	κ=0	κ=lrn	Δ%
Slow	0.0784	**0.0737**	−6.0
Mid	1.0049	**0.8144**	−19.0
Fast	0.2743	**0.2453**	−10.5

Table 3. **Left:** multi-clip beats per-clip on linear phenomena, fails on nonlinear pendulum. **Right:** κ reduces MSE 6–19%, confirming physical signal.

provides the comprehensive C vs. M across all scenarios; MAE=0 (sliding cone) is exact protractor, not fitted, the easy end. **CNN setup:** ResNet-18 (ImageNet-pretrained), 5 sampled frames → mean pool → 8-way softmax; 80/20 split (192/48 clips). "In-distribution" = same 8 phenomena.

[1] ECCV 2026 Program Chairs: Reviewer contribution-type guidelines. Link 1